

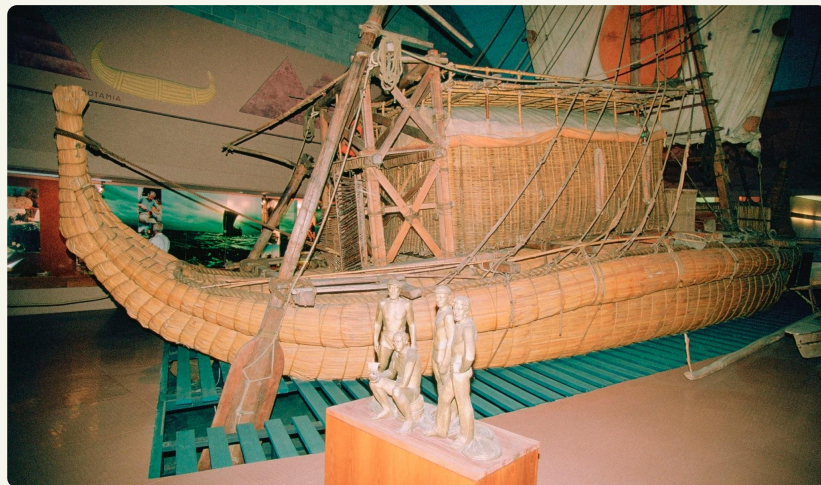
Moisture + Light Proof Bookcase

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Maxwell



Problem

- Important historical material is sensitive to light and humidity
- There are more than 100 million rare books which conservators must protect ([library of congress](#))
- About 85% of museums have a maintenance and repair backlog / flooding concern / lack of climate controlled storage ([Facilities Dive](#))



Solution

- A cheap, easy, compact alternative to large scale climate control projects
- \$ Repairs > \$ Individualized climate control for valuable items
 - Less risk to other items
- Reactive / Protects individual objects



Practical Application:

Museum Collections



([Dubansky 2023](#))

Importation Historical documentation

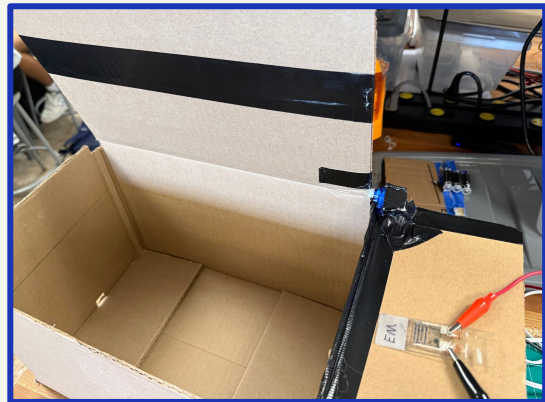
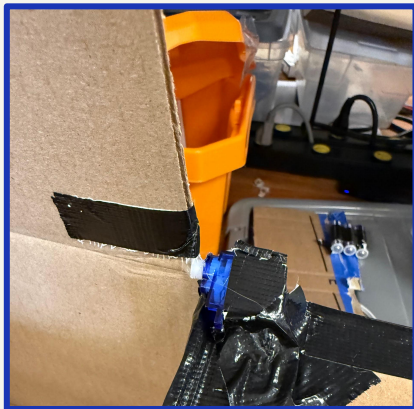
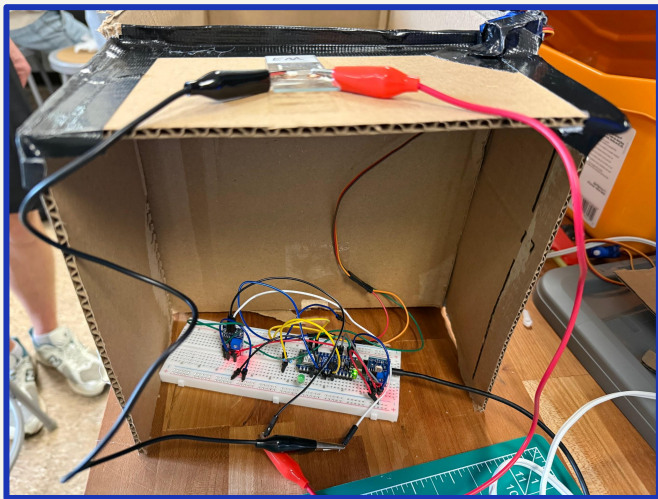
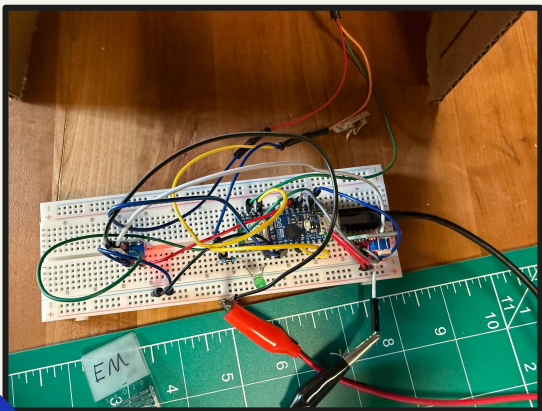
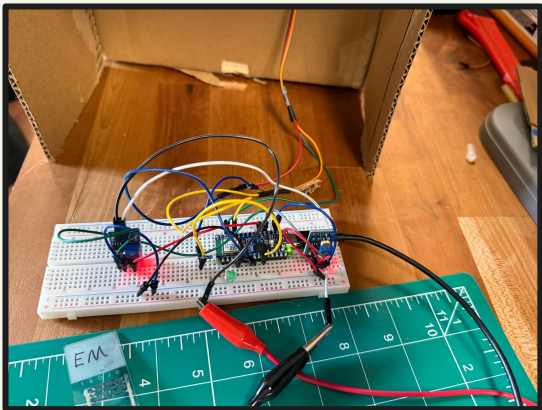


([Getty Images](#))

Personal Collections



([Chesnot/Getty Images](#))



Use of AI

- Copilot was used to generate the micropython code.
- Copilot also gave us the necessary wiring layouts.



Copilot

Video

IMG_3897.MOV

Results/challenges

Light Sensor

Succeeded

- Finicky at first
- Threshold adjustments
- Code editing

Moisture Sensor

Succeeded

- Electronic sensor unresponsive
- Handmade sensor switch

LED

Succeeded

- Wrong PIN coded for (arduino vs. printed)

Servo

Succeeded

- Started with step motor
- Changed to servo

Conclusion



Working Prototype

Successfully incorporates integrated **Water Sensors** and **Light Sensors** into a unified protective monitoring system.



Practical Application

Serves as a proof-of-concept for real-world deployment designed specifically to safeguard **delicate media**.

Questions?